

statsmodels.stats.diagnostic.het_breuschpagan

```
statsmodels.stats.diagnostic.het_breuschpagan(  
    resid,  
    exog_het,  
    robust=True  
)
```

[\[source\]](#)

Breusch-Pagan Lagrange Multiplier test for heteroscedasticity

The tests the hypothesis that the residual variance does not depend on the variables in x in the form

Homoscedasticity implies that $\alpha = 0$.

Parameters

resid : `array_like`

For the Breusch-Pagan test, this should be the residual of a regression. If an array is given in `exog`, then the residuals are calculated by the an OLS regression or `resid` on `exog`. In this case `resid` should contain the dependent variable. `Exog` can be the same as `x`.

exog_het : `array_like`

This contains variables suspected of being related to heteroscedasticity in `resid`.

robust : `bool`, default `True`

Flag indicating whether to use the Koenker version of the test (default) which assumes independent and identically distributed error terms, or the original Breusch-Pagan version which assumes residuals are normally distributed.

Returns

- **lm** (*float*) – lagrange multiplier statistic
- **lm_pvalue** (*float*) – p-value of lagrange multiplier test

- **fvalue** (*float*) – f-statistic of the hypothesis that the error variance does not depend on x
- **f_pvalue** (*float*) – p-value for the f-statistic

Notes

Assumes x contains constant (for counting dof and calculation of R^2). In the general description of LM test, Greene mentions that this test exaggerates the significance of results in small or moderately large samples. In this case the F-statistic is preferable.

Verification

Chisquare test statistic is exactly ($<1e-13$) the same result as `bptest` in R-stats with defaults (`studentize=True`).

Implementation

This is calculated using the generic formula for LM test using R^2 (Greene, section 17.6) and not with the explicit formula (Greene, section 11.4.3), unless `robust` is set to `False`. The degrees of freedom for the p-value assume x is full rank.

References

- [1] Greene, W. H. *Econometric Analysis*. New Jersey. Prentice Hall; 5th edition. (2002).
- [2] Breusch, T. S.; Pagan, A. R. (1979). "A Simple Test for Heteroskedasticity and Random Coefficient Variation". *Econometrica*. 47 (5): 1287–1294.
- [3] Koenker, R. (1981). "A note on studentizing a test for heteroskedasticity". *Journal of Econometrics* 17 (1): 107–112.

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